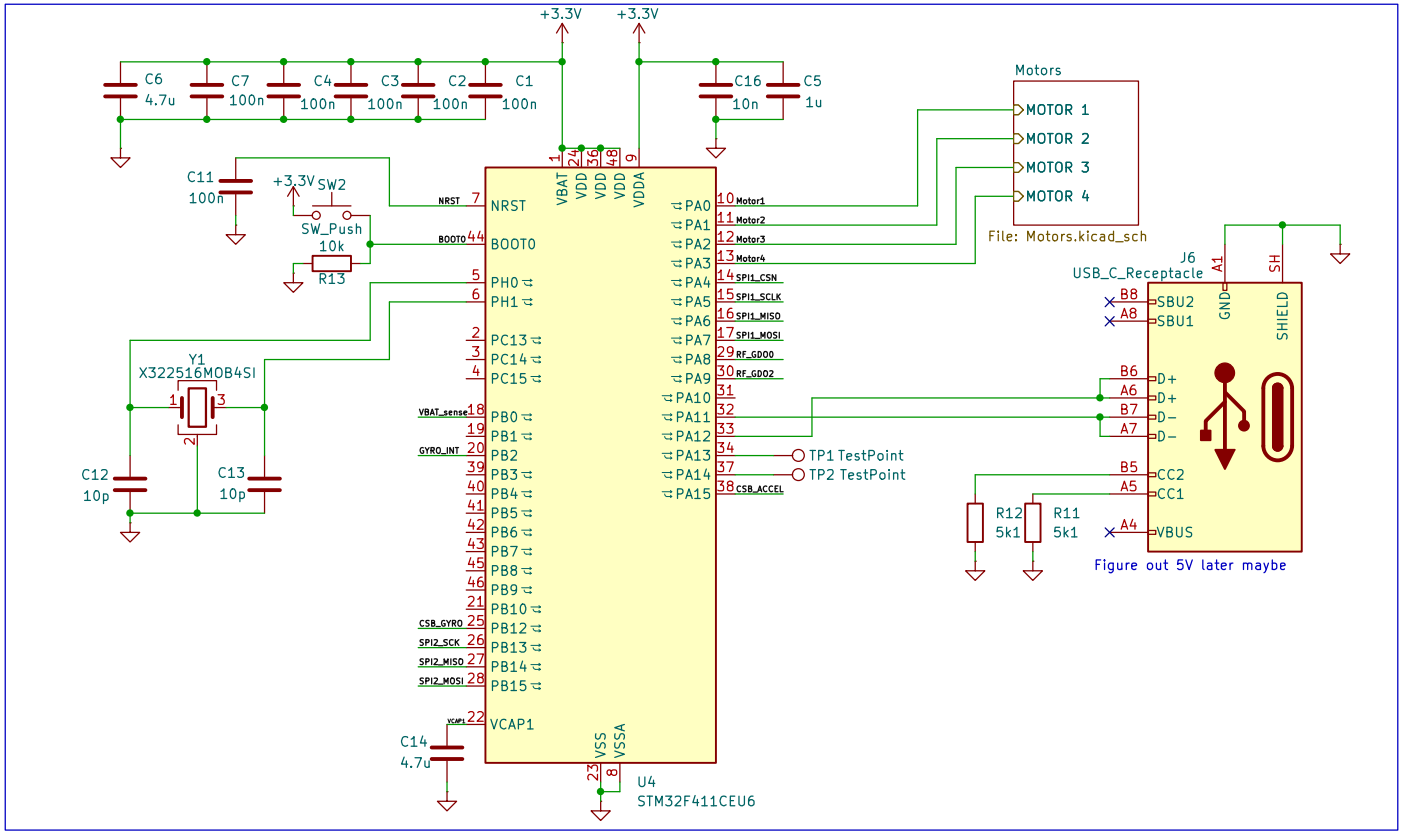
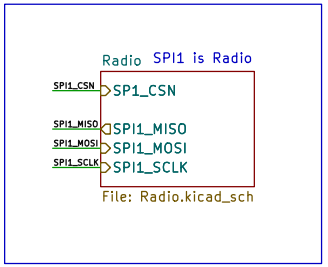
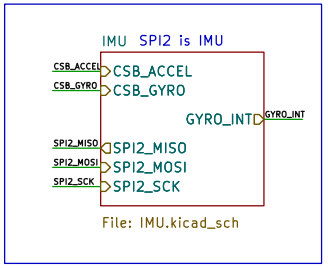
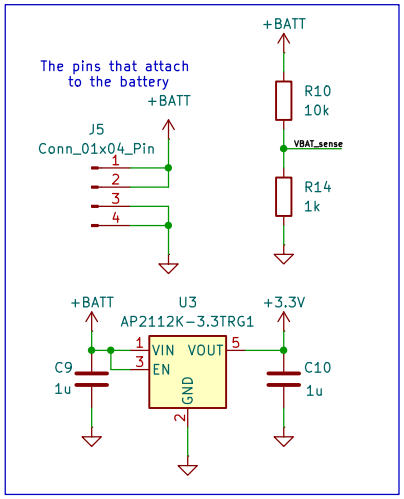
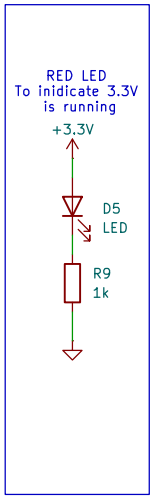
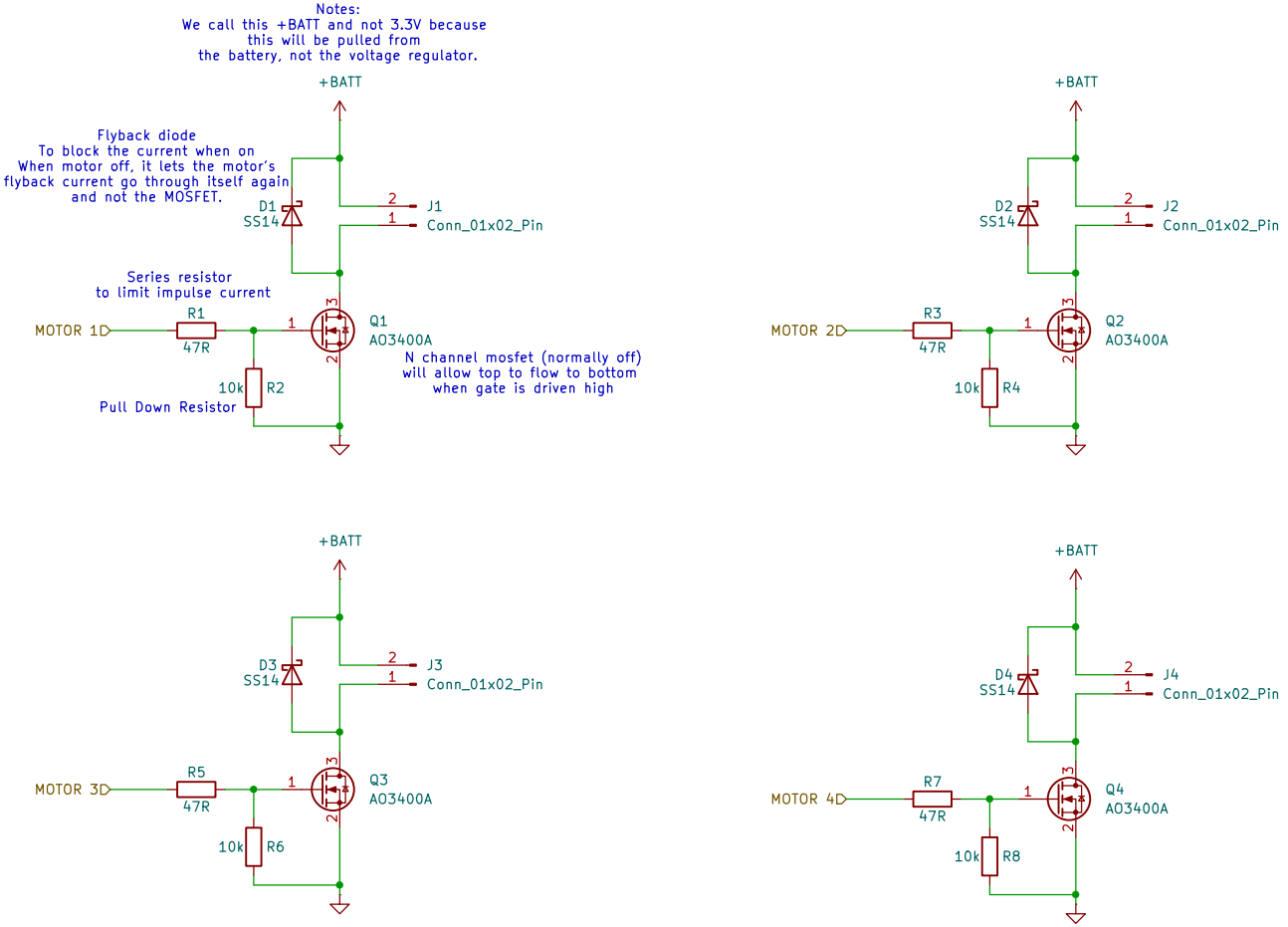

+3.3V
+3.3V
C6
4.7u
C7
100n
C4
100n
C3
100n
C2
100n
C1
100n
C16
10n
C5
1u
C11
100n
+3.3V
SW2
SW_Push
10k
R13
NRST
BOOT0
Y1
X322516MOB4SI
C12
10p
C13
10p
VBAT_sense
GYRO_INT
CSB_GYRO
SPI2_SCK
SPI2_MISO
SPI2_MOSI
VCAP1
C14
4.7u
VBAT
VDD
VDD
VDD
VDDA
VSS
VSSA
NRST
BOOT0
PH0
PH1
PC13
PC14
PC15
PB0
PB1
PB2
PB3
PB4
PB5
PB6
PB7
PB8
PB9
PB10
PB12
PB13
PB14
PB15
VCAP1
PA0
PA1
PA2
PA3
PA4
PA5
PA6
PA7
PA8
PA9
PA10
PA11
PA12
PA13
PA14
PA15
Motor1
Motor2
Motor3
Motor4
SPI1_CSN
SPI1_SCLK
SPI1_MISO
SPI1_MOSI
RF_GDO0
RF_GDO2
CSB_ACCEL
TP1 TestPoint
TP2 TestPoint
U4
STM32F411CEU6
Motors
MOTOR 1
MOTOR 2
MOTOR 3
MOTOR 4
File: Motors.kicad_sch
J6
USB_C_Receptacle
SBU2
SBU1
D+
D+
D−
D−
CC2
CC1
VBUS
GND
SHIELD
R12
5k1
R11
5k1
Figure out 5V later maybe

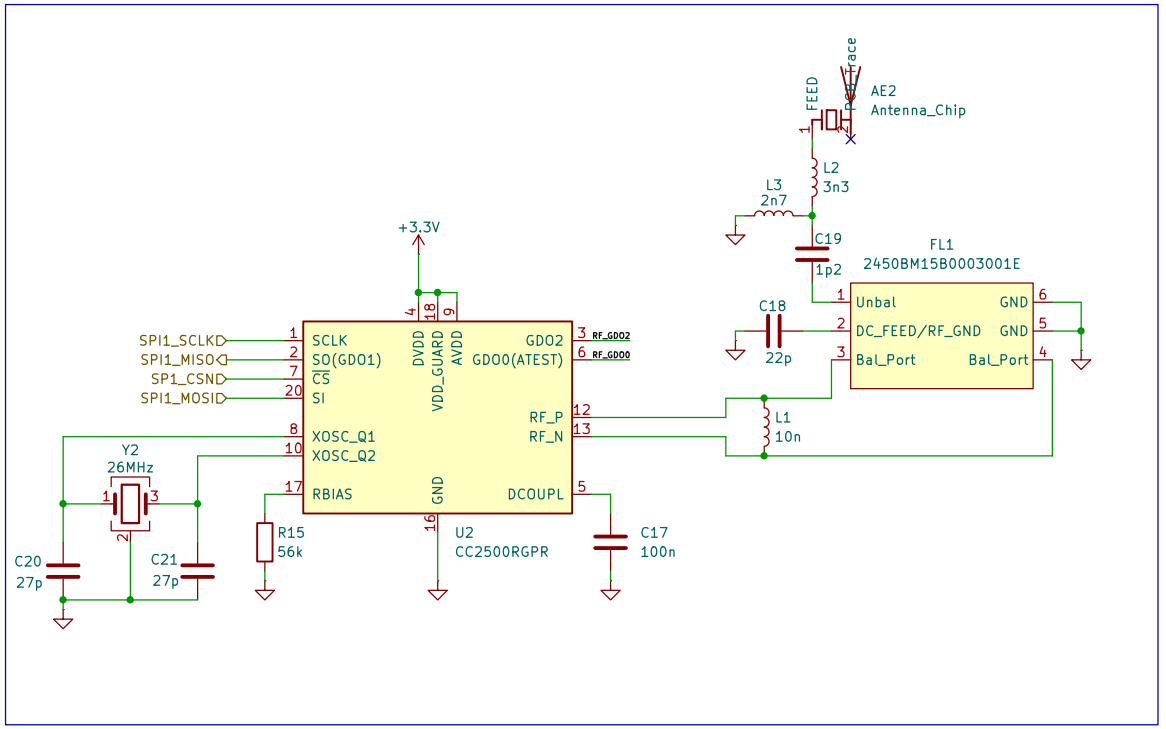

Radio
SPI1 is Radio
SPI1_CSN
SPI1_MISO
SPI1_MOSI
SPI1_SCLK
File: Radio.kicad_sch

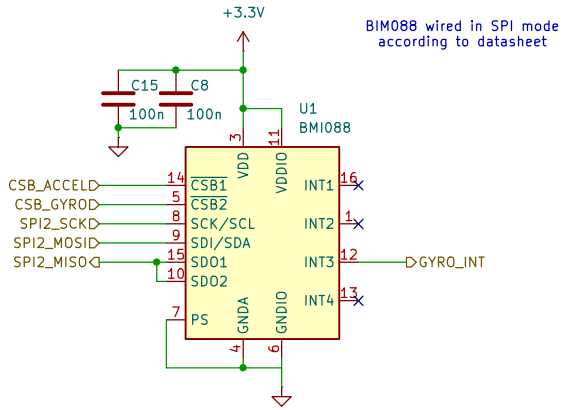

IMU
SPI2 is IMU
CSB_ACCEL
CSB_GYRO
GYRO_INT
SPI2_MISO
SPI2_MOSI
SPI2_SCK
File: IMU.kicad_sch


The pins that attach
to the battery
+BATT
J5
Conn_01x04_Pin
+BATT
R10
10k
VBAT_sense
R14
1k
+BATT
U3
AP2112K-3.3TRG1
+3.3V
VIN
VOUT
EN
GND
C9
1u
C10
1u


RED LED
To inidicate 3.3V
is running
+3.3V
D5
LED
R9
1k


Notes:
We call this +BATT and not 3.3V because
this will be pulled from
the battery, not the voltage regulator.
+BATT
Flyback diode
To block the current when on
When motor off, it lets the motor's
flyback current go through itself again
and not the MOSFET.
D1
SS14
J1
Conn_01x02_Pin
Series resistor
to limit impulse current
MOTOR 1
R1
47R
Q1
AO3400A
N channel mosfet (normally off)
will allow top to flow to bottom
when gate is driven high
10k
R2
Pull Down Resistor
+BATT
D2
SS14
J2
Conn_01x02_Pin
MOTOR 2
R3
47R
Q2
AO3400A
10k
R4
+BATT
D3
SS14
J3
Conn_01x02_Pin
MOTOR 3
R5
47R
Q3
AO3400A
10k
R6
+BATT
D4
SS14
J4
Conn_01x02_Pin
MOTOR 4
R7
47R
Q4
AO3400A
10k
R8


+3.3V
SPI1_SCLK
SPI1_MISO
SP1_CSND
SPI1_MOSI
SCLK
SO(GDO1)
CS
SI
XOSC_Q1
XOSC_Q2
RBIAS
DVDD
VDD_GUARD
AVDD
GND
GDO2
GDO0(ATEST)
RF_P
RF_N
DCOUPL
RF_GDO2
RF_GDO0
U2
CC2500RGPR
Y2
26MHz
C20
27p
C21
27p
R15
56k
C17
100n
L1
10n
C18
22p
C19
1p2
L2
3n3
L3
2n7
FEED
Net-Trace
AE2
Antenna_Chip
FL1
2450BM15B0003001E
Unbal
DC_FEED/RF_GND
Bal_Port
GND
GND
Bal_Port


+3.3V
BIM088 wired in SPI mode
according to datasheet
C15
100n
C8
100n
U1
BMI088
3
11
VDD
VDDIO
CSB_ACCEL
14
CSB1
16
INT1
CSB_GYRO
5
CSB2
SPI2_SCK
8
SCK/SCL
1
INT2
SPI2_MOSI
9
SDI/SDA
SPI2_MISO
15
SDO1
12
INT3
GYRO_INT
10
SDO2
7
PS
GNDA
GNDIO
INT4
13
4
6